

Regime-Dependent Utility of Open-Weight Large Language Models for Tabular Anomaly Detection

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Abstract—Tabular anomaly detection underpins fraud monitoring and intrusion detection. Recent work reports that compact open-weight large language models (LLMs) can be competitive on benchmark AUROC, while other studies find LLMs underperform classical detectors. We conducted a controlled replication and extension of AnoLLM, a recent likelihood-based method. We reproduced SmolLM-360M likelihood scoring on the 30-dataset ODDS anomaly-detection benchmark suite, then compared likelihood scoring (base model with LoRA fine-tuning) against prompted scoring (frozen instruction-tuned model) for matched SmolLM-360M and Qwen2.5-3B models. We further evaluated credit-card fraud and network-intrusion data (UNSW-NB15) under a fixed 1% false-positive-rate (FPR) alert budget, along with several additional ablations. The replication passed the two pre-registered aggregate gates (mean AUROC 0.851 vs. 0.865 published; rank correlation 0.875) but failed the stricter per-dataset band gate (19 of 30 datasets, threshold 24). On ODDS, likelihood scoring dominated prompted scoring (Friedman omnibus, p less than $1e-11$), and the larger Qwen2.5-3B model did not significantly outperform SmolLM-360M under likelihood scoring. On credit-card fraud and UNSW-NB15, recall at the 1% FPR budget favored classical detectors over the tested LLM configurations (e.g., KNN reached 0.715 versus 0.136 for Qwen2.5-3B likelihood scoring on credit-card fraud). Domain-informed column ordering and two-stage triage did not improve results. Open-weight LLM likelihood scoring remains competitive at the aggregate benchmark level, but the tested methods do not outperform classical detectors under fixed-alert-budget conditions on the security-relevant data evaluated here. Practitioners should not treat ODDS-style AUROC as a proxy for deployment-threshold recall.

Index Terms—Anomaly detection, large language models, replication studies, security analytics, tabular data

I. INTRODUCTION

Tabular anomaly detection supports fraud monitoring and intrusion detection. Classical detectors such as Isolation Forest (IForest), k -nearest neighbors (k NN), and ECOD remain strong, low-cost baselines [1], [2].

Recent work asks whether large language models (LLMs) can detect anomalies from serialized tabular rows. AnoLLM [3] showed that small open models scoring rows by likelihood are competitive on ODDS, a standard suite of 30 tabular anomaly-detection benchmark datasets, under AUROC, while AD-LLM [4] reports that contemporary LLMs can underperform classical methods in NLP anomaly-detection settings, a different task domain.

The field therefore faces a precise, unresolved question: under what operating regime—model scale, scoring mode, data semantics, imbalance, and alert budget—do open-weight LLMs help for tabular anomaly detection, and where do classical detectors still dominate? These conflicting findings suggest that LLM utility may depend less on model family alone than on the scoring rule, benchmark, and operating metric.

We study this question through a replication and extension of AnoLLM rather than by proposing a new detector. Our study holds model family and size fixed while comparing two scoring modes: (A) base model plus Low-Rank Adaptation (LoRA) fine-tuned likelihood scoring versus (B) a frozen instruction-tuned sibling scored by prompted expected value. We pre-registered replication gates before generating results, then extended evaluation to two security-relevant datasets with fixed-false-positive-rate (FPR) operational metrics, serialization ablations, and a two-stage triage design.

Our contributions are threefold. First, we provide a controlled same-family, same-size likelihood-versus-prompted A/B on open-weight models (SmolLM-360M and Qwen2.5-3B); the base-versus-instruct checkpoint difference between the two modes is bounded empirically in Section III (mean $|\Delta\text{AUROC}| = 0.0054$). Second, we re-evaluate under realistic operating conditions on credit-card fraud and UNSW-NB15 using recall@1%FPR, plus semantic-name and serialization-order ablations. Third, we report three negative findings: the tested domain-informed column ordering did not help on pima and UNSW-NB15; semantic column names did not reliably help on pima ($n = 3$ pairs, exploratory); and IForest-plus-LLM triage added zero uplift at $k = 1\%$.

The security experiments rest on two datasets and are reported as per-dataset evidence only. ODDS results use AUROC; security results use recall@1%FPR. The resulting evidence should therefore be interpreted as a replication and evaluation study rather than as a new detector proposal. Overall, the results support a regime-dependent interpretation: likelihood scoring remains competitive under ODDS AUROC, whereas the tested open-weight configurations are not competitive for fixed-FPR alerting on the two security-relevant datasets. Table I summarizes the evaluation regimes and supported claims; every row is scoped to the datasets,

metrics, and methods actually tested. The remainder of this paper reviews related work (Section II), describes the experimental method (Section III), reports replication, scoring-mode, security-transfer, and ablation results (Section IV), discusses implications, limitations, and future directions (Section V), and concludes (Section VI).

II. RELATED WORK

LLMs for tabular anomaly detection. AnoLLM [3] fine-tunes SmolLM models and scores rows by mean negative log-likelihood over random column permutations, reporting competitiveness with classical detectors on ODDS under AUROC. It does not run a same-family likelihood-versus-prompted comparison, does not sweep open-weight scale for prompted scoring, and evaluates primarily by AUROC rather than fixed-FPR metrics. AnoLLM’s mixed-type evaluation includes two fraud-related datasets (fraud-ecommerce, vehicle-insurance) at 6–9% anomaly rates; our security evaluation targets a substantially more imbalanced regime (0.17% on credit-card fraud) using fixed-FPR metrics not reported in the original study. Li et al. [5] study *batch-level* tabular anomaly detection, where the model sees a batch of rows and identifies the outlier row; they report GPT-4 competitive with state-of-the-art transductive methods on ODDS. Our work differs in three ways: scoring is *per-row* (not batch-level), models are open-weight (not GPT-4), and we run a controlled A/B comparing likelihood to prompted scoring. AD-LLM [4] benchmarks LLMs for anomaly detection in NLP settings (zero-shot text detection, data augmentation, model selection); it does not study tabular anomaly detection directly. Its findings motivate our regime-dependent framing, although it is not a direct tabular benchmark.

Tabular serialization and generative tabular models. GReaT [6] and TabLLM [7] serialize rows to text for generation or classification. We reuse row serialization but study anomaly *scoring* and treat column order as an explicit variable. CausalTAD [8] proposes causal column ordering (modeled as a linear ordering problem) plus causal-aware column reweighting, arguing that causal structure between columns affects likelihood. Our serialization-order ablation addresses a strictly narrower question: whether a *manually defined domain-informed order* improves prompted scoring. We do not test causal discovery, causal ordering derived from data, or column reweighting; a null result there does not address CausalTAD’s causal reweighting mechanism.

Benchmarks and classical baselines. ADBench [2] and the ODDS library standardize datasets and baseline panels. We adopt IForest, PCA, k NN, and ECOD from PyOD [1] for comparability with AnoLLM’s classical panel. To our knowledge, no prior work jointly provides (i) a controlled same-family, same-size open-weight likelihood-versus-prompted A/B with an empirical checkpoint bound, (ii) fixed-FPR evaluation on two security-relevant datasets, and (iii) a reported negative two-stage triage result. These distinctions position the paper as a replication and evaluation extension rather than a new detection method.

III. METHOD

A. Row Serialization

Let a row x have d columns with names $(c_1, \dots, c_d)$ and values $(v_1, \dots, v_d)$. Each row was rendered as `col is value, col is value, ...` following AnoLLM. Numerical preprocessing on ODDS followed AnoLLM’s “standard” protocol; the same transformation was applied consistently across likelihood and prompted comparisons. Security loaders returned raw floats; this protocol difference is documented and bounded by a binned credit-card check (BA1). The same serialization was shared across both scoring modes within each comparison. Column order was fixed within a comparison and varied only in the serialization-order ablation.

B. Scoring Modes

Mode A (likelihood). The base model (SmolLM-360M or Qwen2.5-3B) was fine-tuned with LoRA [9] on normal training rows. For a test row x and permutation r , let $T_r(x)$ denote the token positions corresponding to serialized field values (column-name tokens excluded). The anomaly score was

$$s(x) = \frac{1}{R} \sum_{r=1}^R \frac{1}{|T_r(x)|} \sum_{t \in T_r(x)} -\log p_\theta(y_t | y_{<t}), \quad (1)$$

averaged over $R = 10$ random column permutations, with per-column length normalization for textual fields.

Mode B (prompted). The frozen instruction-tuned sibling (SmolLM-360M-Instruct or Qwen2.5-3B-Instruct) read the serialized row. Prompts used the form: “Record: [serialized row]. How anomalous is this record on a scale of 0 (normal) to 9 (highly anomalous)? Answer with a single digit.” We computed a continuous score as the expected value over anomaly-level digit tokens from a single forward pass. This expected-value scoring is one specific instantiation of prompted anomaly detection. We do not evaluate few-shot, chain-of-thought, pairwise, or calibrated temperature-scaled variants. Conclusions on Mode B apply to this baseline only and should not be taken as the final word on all prompting methods.

Model choice and scale range. SmolLM-360M [10] was chosen to match AnoLLM’s backbone for the replication. Qwen2.5-3B [11] was chosen as an open-weight model at an $8\times$ larger scale feasible within the same hardware budget. The study is limited to the 360M–3B parameter range; results may not hold for 7B-scale models, different architectures, or instruction-tuned checkpoints outside the tested SmolLM-360M and Qwen2.5-3B families.

Mode A uses a base+LoRA checkpoint and Mode B uses a frozen instruct checkpoint, so the two modes differ in both scoring method and checkpoint. A checkpoint-confound control (DA1) LoRA-fine-tuned Qwen2.5-3B-Instruct and scored by likelihood on eight ODDS datasets, comparing against the corresponding Qwen2.5-3B base+LoRA cells, to bound this confound.

TABLE I
SUMMARY OF EVALUATION REGIMES AND SUPPORTED CLAIMS

Regime	Metric	Finding (tested methods)
ODDS (30 datasets)	AUROC	Likelihood competitive at aggregate level; prompted expected-value baseline underperforms
ODDS (30 datasets)	AUROC	No significant difference between SmoLLM-360M and Qwen2.5-3B under likelihood scoring
Credit-card + UNSW	Recall@1%FPR	Classical detectors outperform tested LLM configurations (prompted only on UNSW)
Credit-card only	Recall@1%FPR	Qwen likelihood scoring underperforms the classical panel (likelihood not run on UNSW by design)
Ablations	AUROC	Domain ordering: null on pima and UNSW-NB15; semantic names: null on pima only
Triage	Recall@FPR uplift	Zero uplift at $k=1\%$; harmful at wider k

C. Metrics and Statistics

AUROC (tie-aware) was used for ODDS comparability; it measures ranking quality across all score thresholds and does not specify performance at a single alert budget. On security data we reported recall@1%FPR as the primary metric and area under the precision-recall curve (AUPRC) relative to prevalence with importance reweighting after negative subsampling as a secondary imbalance-aware view. Recall@1%FPR is the recall at the ROC operating point τ^* with the largest false-positive rate not exceeding 1%, computed from scikit-learn’s ROC curve (ties grouped at shared thresholds).

$$\text{Recall@1\%FPR} = \text{TPR}(\tau^*). \quad (2)$$

Recall@1%FPR is the primary security metric because operational fraud and intrusion-detection systems typically work at a fixed alert budget: only the top- k ranked cases are investigated. ODDS ($n = 30$ datasets): Friedman omnibus, Nemenyi critical-difference diagram, and Holm-corrected Wilcoxon signed-rank for pre-registered pairwise claims. Security ($n = 2$ datasets): per-dataset mean $\pm$ std over 3 seeds; no Friedman test.

D. Experimental Controls

Replication gates (C1–C3), confound checks (DA1, BA1), experiment grids, AnoLLM reference targets, and mode-specific hyperparameters were fixed before result generation. Prompt templates for Mode B were held constant across runs. Each completed cell recorded seeds, train/test splits, hyperparameters, and metrics in standardized run metadata. Likelihood runs used SmoLLM-360M at 2000 training steps and Qwen2.5-3B at 1000 steps, with $R = 10$ random column permutations and LoRA fine-tuning via the Parameter-Efficient Fine-Tuning (PEFT) library. Dependencies were pinned to torch==2.3.1 and transformers==4.48.2. We do not claim bit-exact reproduction of AnoLLM’s published fork; the per-dataset shortfall is reported in the replication results.

E. Protocol Checks

DA1 (checkpoint-confound control). Instruct-checkpoint likelihood yielded mean $|\Delta\text{AUROC}| = 0.0054$ vs.

base+LoRA across eight ODDS datasets (threshold 0.02). **BA1 (binned credit-card).** Switching credit-card from raw floats to ODDS-style binning changed mean classical AUROC by 0.0012 across four detectors (threshold 0.03). **T3 (Time feature).** On temporal credit-card splits, including the Time column collapsed k NN AUROC from 0.932 to 0.178. The Time column encodes absolute transaction order; including it introduces a non-stationarity artifact that can dominate distance-based scores, so all reported classical credit-card results use the time-excluded protocol.

IV. RESULTS

A. Replication

SmoLLM-360M likelihood scoring passed the two pre-registered aggregate gates (C1, C2) but failed the stricter per-dataset band gate (C3). Mean AUROC was 0.851 vs. AnoLLM’s reported 0.865 ($\Delta = 0.0145$, C1 pass). Spearman rank correlation across 30 datasets was $\rho = 0.875$ (C2 pass). Nineteen of 30 datasets fell within the pre-registered per-dataset band (C3 threshold 24/30, fail). Under the pre-registered decision rule, C1 and C2 were the hard-stop criteria; the replication is a credible partial reproduction. These checks did not identify a protocol error explaining the per-dataset deviations.

B. Scoring Mode

Likelihood scoring dominated prompted scoring across 360 cells (30 ODDS datasets $\times$ 4 conditions (2 models $\times$ 2 modes) $\times$ 3 seeds). The Friedman omnibus test rejected equal performance ($\chi_F^2 = 55.27$, $p = 6.0 \times 10^{-12}$, $k = 4$, $n = 30$). Average ranks (lower is better): SmoLLM-likelihood 1.62, Qwen-likelihood 1.65, SmoLLM-prompted 3.20, Qwen-prompted 3.53. A likelihood condition ranked first on 28 of 30 datasets; both prompted conditions ranked third or fourth on 28 of 30. Holm-corrected Wilcoxon tests against SmoLLM-likelihood: both prompted conditions were significantly worse (SmoLLM-prompted: median $\Delta = -0.276$, $p_{\text{Holm}} = 1.4 \times 10^{-7}$; Qwen-prompted: median $\Delta = -0.339$,

$p_{\text{Holm}} = 9.4 \times 10^{-7}$). Thus, for ODDS-style AUROC, likelihood scoring rather than prompted expected-value scoring is the effective LLM-based scoring regime in our experiments.

Table II summarizes condition means. Fig. 1 shows the critical-difference diagram, where the two likelihood conditions form a cluster distinct from both prompted conditions. The replication run ($\text{max_steps}=2000$) reports SmolLM-360M AUROC 0.851; the scoring-mode comparison uses independently initialized runs, giving 0.843 for the same model and mode. The difference (0.008) reflects run-to-run variance across independently seeded training runs; the C1 replication gate (Section IV-A) is evaluated on the pre-registered replication run (0.851) and is not affected by this later, independently seeded comparison.

TABLE II
ODDS MEAN AUROC AND AVERAGE RANK BY CONDITION ($n = 30$ DATASETS, 3 SEEDS)

Condition	Mean AUROC	Avg. rank
SmolLM-360M / likelihood	0.843	1.62
Qwen2.5-3B / likelihood	0.859	1.65
SmolLM-360M / prompted	0.610	3.20
Qwen2.5-3B / prompted	0.493	3.53

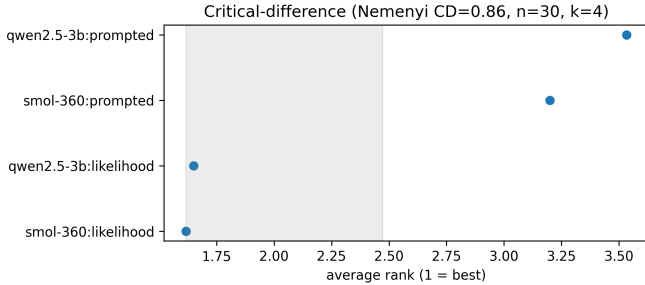


Fig. 1. Critical-difference diagram for four conditions on 30 ODDS datasets (Friedman $p = 6.0 \times 10^{-12}$). Connected conditions are not significantly different by Nemenyi post-hoc test.

C. Model Scale

Within likelihood scoring, Qwen2.5-3B (mean AUROC 0.859) did not significantly outperform SmolLM-360M (0.843; Holm-corrected Wilcoxon $p = 0.77$). This is consistent with AnOLLM’s finding that larger models do not improve NLL-based anomaly scoring on predominantly numerical ODDS data. Scale upgrades within the tested open-weight range do not justify the added compute for likelihood scoring on ODDS.

D. Semantic Column Names

On pima, replacing integer column indices with descriptive names changed mean AUROC by -0.007 ± 0.021 across three seed pairs (individual deltas: $+0.016$, -0.026 , -0.010). This is a null result at $n = 3$ pairs (minimum achievable Wilcoxon $p = 0.25$); we report descriptive evidence only. Semantic

column names did not reliably improve prompted or likelihood scoring at this granularity.

E. Fixed-FPR Security Evaluation

Across the completed security-relevant evaluations, classical detectors outperformed the tested LLM configurations at the 1% FPR operating point. Table III reports $\text{recall@1\%FPR}$ (mean $\pm$ std over 3 seeds). Likelihood scoring (mode A) was run on credit-card fraud only (6 cells: 2 splits $\times$ 3 seeds); UNSW-NB15 likelihood was omitted by pre-specified design to bound GPU cost. SmolLM-360M likelihood scoring was not included in the pre-specified security grid; the likelihood arm targets Qwen2.5-3B only, consistent with the pre-registered design. UNSW LLM comparisons therefore use prompted scoring only. On credit-card fraud (random split, time excluded), KNN achieved 0.715 ± 0.015 , ECOD 0.680 ± 0.003 , and IForest 0.586 ± 0.029 , while Qwen likelihood reached 0.136 ± 0.017 and Qwen prompted 0.003 ± 0.001 . On UNSW-NB15, the leading classical detectors (KNN: 0.189 ± 0.012 , IForest: 0.188 ± 0.018) outperformed Qwen prompted (0.151 ± 0.013). The gap was large on credit-card fraud and smaller but still positive for the best classical baseline on UNSW-NB15. AUPRC-gain confirmed the same ordering (ECOD: $88\times$, IForest: $66\times$, KNN: $59\times$, PCA: $76\times$ on credit-card fraud versus Qwen likelihood: $7\times$, prompted: $\approx 1\times$); classical detectors similarly exceeded LLMs on UNSW-NB15. Fig. 2 plots $\text{recall@1\%FPR}$ by method on both datasets, where the classical detectors cluster well above both Qwen configurations on credit-card fraud, and the gap narrows, but does not close, on UNSW-NB15. AUROC alone is misleading in these settings: IForest AUROC was 0.951 on the credit-card random split, but $\text{recall@1\%FPR}$ was 0.586 vs. 0.136 for Qwen likelihood; ECOD AUROC was 0.950 yet $\text{recall@1\%FPR}$ was 0.680. Section IV-H examines whether two-stage re-ranking closes this gap.

TABLE III
RECALL@1%FPR ON TWO SECURITY-RELEVANT DATASETS (MEAN $\pm$ STD, 3 SEEDS). CREDIT-CARD USES TIME-EXCLUDED PROTOCOL. † QWEN LIKELIHOOD EVALUATED ON CREDIT-CARD ONLY (PRE-SPECIFIED COMPUTE BUDGET).

Method	CC (random)	UNSW-NB15
KNN (classical)	0.715 ± 0.015	0.189 ± 0.012
ECOD (classical)	0.680 ± 0.003	0.161 ± 0.006
PCA (classical)	0.631 ± 0.004	0.099 ± 0.002
IForest (classical)	0.586 ± 0.029	0.188 ± 0.018
Qwen2.5-3B / likelihood	0.136 ± 0.017	† —
Qwen2.5-3B / prompted	0.003 ± 0.001	0.151 ± 0.013
SmolLM-360M / prompted	0.011 ± 0.004	0.009 ± 0.001

F. Serialization Order

The domain ordering for UNSW-NB15 placed network-level identification features (e.g., source/destination ports, protocol type) first, followed by traffic volume statistics,

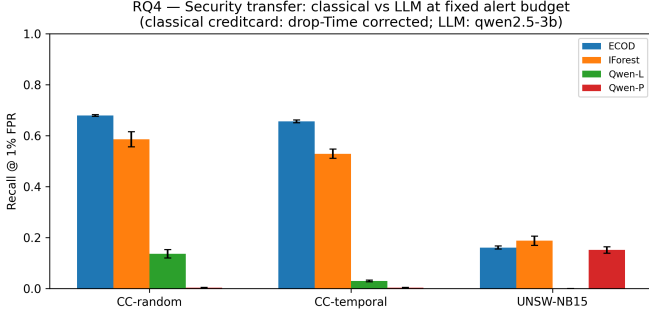


Fig. 2. Recall@1%FPR by method on credit-card fraud and UNSW-NB15.

then payload features. For pima it followed demographic and clinical measurement fields in the dataset’s conventional order. Both orderings were defined prior to running the experiment and held fixed across all seeds. Mean AUROC by ordering condition (Qwen2.5-3B prompted, 3 seeds) is shown in Fig. 3. Pooled across pima and UNSW-NB15: arbitrary 0.564, random-0 0.550, random-1 0.543, domain 0.501. On pima (8 columns), arbitrary and domain orderings tied at 0.449 AUROC; random column shuffles achieved 0.526 and 0.553, indicating the fixed arbitrary ordering is suboptimal for this dataset. On UNSW-NB15 (47 columns), arbitrary ordering (0.680) outperformed domain ordering (0.554) by 0.126 AUROC. The tested domain-informed ordering did not improve LLM anomaly scoring on pima (tied with arbitrary at 0.449) and hurt performance on UNSW-NB15 relative to the default arbitrary ordering.

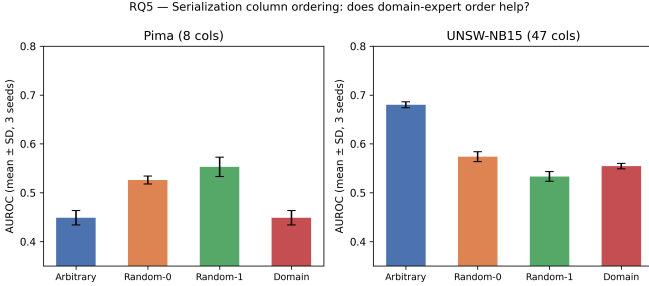


Fig. 3. Mean AUROC by serialization column ordering on pima and UNSW-NB15 (Qwen2.5-3B prompted, 3 seeds per condition).

G. Practicality

Fig. 4 plots mean AUROC against wall time per 1,000 test rows. Timings were measured as pod-level elapsed runtime on RunPod A40 GPUs within the same hardware pool; they include LoRA fine-tuning and inference for Mode A, and inference only for Mode B, but not dataset download. Qwen2.5-3B likelihood achieved mean AUROC 0.859 at ≈ 222 s per 1,000 test rows (pod-level approximation for the likelihood runs). Qwen2.5-3B prompted achieved 0.493 at ≈ 33 s per 1,000 test rows. IForest runs in under 5 s per 1,000 test rows with AUROC above 0.80 on ODDS. The Pareto figure also shows

a triage point (0.687 AUROC at ≈ 54 s per 1,000 test rows); this is the LLM re-ranker component’s AUROC evaluated in isolation, not the two-stage system AUROC. The full triage system does not improve over IForest on the security-relevant datasets; its utility is evaluated by recall@FPR uplift in Section IV-H. Likelihood scoring offers higher AUROC at substantially higher cost than prompted scoring; in this implementation, IForest occupies a more favorable speed-accuracy region than the tested LLM scoring modes.

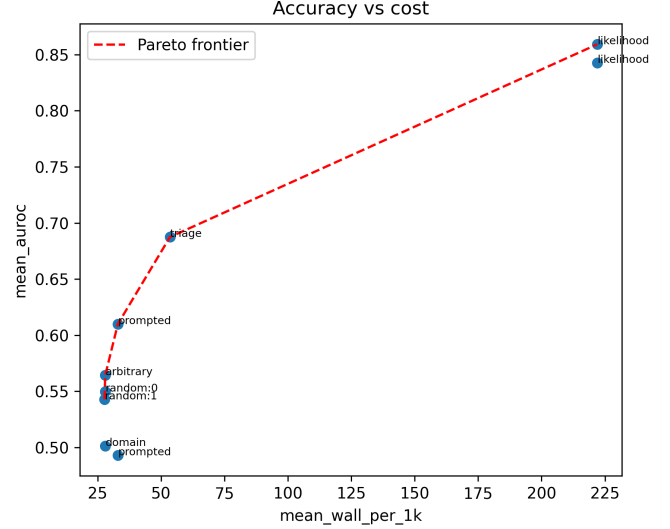


Fig. 4. Mean AUROC vs. wall time per 1,000 test rows for tested LLM and classical scoring modes.

H. Two-Stage Triage

Two-stage triage re-ranks an IForest shortlist using Qwen2.5-3B prompted scoring; its effect is measured as an uplift in recall@ k FPR relative to IForest alone:

$$\text{Uplift}(k) = \text{Recall}@k\text{FPR}_{\text{triage}} - \text{Recall}@k\text{FPR}_{\text{IForest}}. \quad (3)$$

At $k = 1\%$ (IForest shortlist then Qwen2.5-3B prompted re-rank), uplift in recall@1%FPR was 0.00 across all nine cells (3 datasets $\times$ 3 seeds). At $k = 10\%$, the LLM re-ranker was harmful: mean uplift was -0.489 on credit-card random, -0.455 on credit-card temporal, and -0.089 on UNSW-NB15. Table IV summarizes triage uplift. At $k = 1\%$, LLM re-ranking does not recover additional anomalies beyond the IForest shortlist; at wider shortlists, LLM re-ranking lowers recall relative to the original IForest ranking.

V. DISCUSSION

A. Interpretation

The experimental sequence forms a cumulative argument. Replication establishes aggregate agreement with AnOLLM. The scoring-mode and model-scale comparisons show that likelihood scoring, rather than model scale, explains the ODDS result in the tested range. The fixed-FPR security evaluation

TABLE IV
TWO-STAGE TRIAGE UPLIFT IN RECALL@FPR (IForest + QWEN2.5-3B
PROMPTED RE-RANK, MEAN OVER 3 SEEDS). NEGATIVE VALUES
INDICATE LLM RE-RANKING DEGRADED PERFORMANCE.^a

Dataset	$k = 1\%$	$k = 5\%$	$k = 10\%$
Credit-card (random)	0.00	-0.330	-0.489
Credit-card (temporal)	0.00	-0.299	-0.455
UNSW-NB15	0.00	-0.032	-0.089

^aIForest baseline here (0.603) is from the triage experiment; Table III reports 0.586 from an independent security experiment run. Seed variance accounts for the difference.

shows that ODDS AUROC success does not imply deployment utility at fixed alert quotas. The serialization and triage ablations show that natural modifications—semantic names, manual domain-informed ordering, and hybrid triage—do not close the gap.

The ODDS experiments indicate that scoring mode matters more than model scale in the tested range. A likelihood condition ranked first on 28 of 30 datasets; both prompted conditions ranked third or fourth on 28 of 30. Prompted expected-value scoring collapses per-token probability into a single digit, discarding the distributional signal that likelihood scoring preserves.

Across the security-relevant evaluations completed here, classical detectors outperform the tested LLM configurations at fixed 1% FPR. On credit-card fraud, KNN, ECOD, and IForest delivered recall@1%FPR of 0.715, 0.680, and 0.586 while Qwen likelihood reached 0.136 and Qwen prompted 0.003; the likelihood-versus-security comparison is available for credit-card fraud only (UNSW-NB15 likelihood was omitted by pre-specified design). On UNSW-NB15, prompted-only LLM comparisons also show classical detectors ahead. Practitioners operating at a fixed alert quota should evaluate recall at the alert threshold, not AUROC alone.

The tested serialization changes did not improve performance. The tested domain-informed ordering did not help (pima tied at 0.449; UNSW arbitrary exceeded domain by 0.126 AUROC). Semantic column names did not reliably help (mean $\Delta = -0.007$, $n = 3$ pairs).

Two-stage triage does not close the fixed-FPR gap on the two security-relevant datasets. At $k = 1\%$, LLM re-ranking does not recover additional anomalies beyond the IForest shortlist; at wider k , re-ranking is harmful.

AnoLLM [3] reports strong LLM performance under likelihood scoring on tabular ODDS benchmarks; Li et al. [5] show GPT-4-level batch detection competitive with transductive methods on the same data; AD-LLM [4] finds variable LLM performance in NLP anomaly-detection settings. These results are mutually consistent once scoring mode (likelihood vs. prompted), evaluation metric (AUROC vs. recall@FPR), and data regime (benchmark numerical ODDS vs. extreme-imbalance security-relevant data) are stated explicitly. AUROC-only comparisons can therefore be misleading for alert-budgeted deployments on imbalanced data.

B. Limitations

The C3 per-dataset band criterion failed (19/30 vs. threshold 24/30). Timing estimates for likelihood and prompted modes are pod-level approximations, not per-cell wall clock. Security results rest on two datasets; we make no cross-dataset generalization claim. The likelihood-versus-prompted A/B compares base+LoRA vs. frozen instruct; the checkpoint-confound control bounds but does not eliminate this confound. Semantic-name and serialization-order ablations use $n = 3$ seed pairs and are exploratory. We report implementation controls and run metadata sufficient to interpret the experiments, but do not claim artifact-level reproducibility beyond the protocol described here.

C. Future Work

A broader security benchmark (more than five datasets) would enable Friedman-powered cross-dataset tests. Extending likelihood scoring to UNSW-NB15 would clarify whether the credit-card likelihood result also holds in the second security-relevant evaluation. LoRA fine-tuning stability on small tabular datasets warrants investigation given C3 shortfall. Higher-recall shortlisting stages and calibrated prompted scoring remain unexplored alternatives to the tested expected-value baseline.

VI. CONCLUSION

We presented a controlled replication and extension of AnoLLM for open-weight LLM-based tabular anomaly detection, holding model family and size fixed across likelihood and prompted scoring modes. The evidence is regime-dependent: likelihood scoring remains competitive under aggregate ODDS AUROC, matching AnoLLM’s published result on the two pre-registered aggregate gates (the per-dataset band gate failed), but the tested configurations do not outperform classical detectors at fixed alert budgets on the two security-relevant datasets evaluated, and none of the tested modifications (semantic column names, domain-informed ordering, or two-stage triage) closed that gap. Benchmark AUROC and deployment-relevant recall are not interchangeable; the choice between an LLM-based and a classical detector should be made against the metric and regime a system will actually operate under.

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REFERENCES

- [1] Y. Zhao, Z. Nasrullah, and Z. Li, “PyOD: A python toolbox for scalable outlier detection,” *Journal of Machine Learning Research*, vol. 20, no. 96, pp. 1–7, 2019. [Online]. Available: <https://jmlr.org/papers/v20/19-011.html>

- [2] S. Han, X. Hu, H. Huang, M. Jiang, and Y. Zhao, "ADBench: Anomaly detection benchmark," in *Advances in Neural Information Processing Systems*, 2022. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/hash/cf93972b116ca5268827d575f2cc226b-Abstract-Datasets_and_Benchmarks.html
- [3] C.-P. Tsai, G. Teng, P. Wallis, and W. Ding, "AnoLLM: Large Language Models for Tabular Anomaly Detection," in *International Conference on Learning Representations*, 2025. [Online]. Available: <https://openreview.net/forum?id=7VkhffT5X2>
- [4] T. Yang, Y. Nian, L. Li, R. Xu, Y. Li, J. Li, Z. Xiao, X. Hu, R. A. Rossi, K. Ding, X. Hu, and Y. Zhao, "AD-LLM: Benchmarking large language models for anomaly detection," in *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics, 2025, pp. 1524–1547. [Online]. Available: <https://aclanthology.org/2025.findings-acl.79/>
- [5] A. Li, Y. Zhao, C. Qiu, M. Kloft, P. Smyth, M. Rudolph, and S. Mandt, "Anomaly detection of tabular data using LLMs," *arXiv preprint arXiv:2406.16308*, 2024. [Online]. Available: <https://arxiv.org/abs/2406.16308>
- [6] V. Borisov, K. Seßler, T. Leemann, M. Pawelczyk, and G. Kasneci, "Language models are realistic tabular data generators," in *International Conference on Learning Representations*, 2023. [Online]. Available: <https://openreview.net/forum?id=cEygmQNOeI>
- [7] S. Hegselmann, A. Buendia, H. Lang, M. Agrawal, X. Jiang, and D. Sontag, "TabLLM: Few-shot classification of tabular data with large language models," in *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 206. PMLR, 2023, pp. 5549–5581. [Online]. Available: <https://proceedings.mlr.press/v206/hegselmann23a.html>
- [8] R. Wang, R. Liu, R. Chen, H. Suo, Z. Peng, Z. Tang, and C. Chen, "CausalTAD: Injecting causal knowledge into large language models for tabular anomaly detection," *arXiv preprint arXiv:2602.07798*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.07798>
- [9] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations*, 2022. [Online]. Available: <https://openreview.net/forum?id=nZeVKeeFYf9>
- [10] L. Ben Allal, A. Lozhkov, and E. Bakouch, "SmolLM - blazingly fast and remarkably powerful," Hugging Face Blog, 2024. [Online]. Available: <https://huggingface.co/blog/smollm>
- [11] Q. Team, "Qwen2.5 technical report," *arXiv preprint arXiv:2412.15115*, 2025. [Online]. Available: <https://arxiv.org/abs/2412.15115>